

Project Development Phase

Model Performance Test (Artificial Intelligence)

Date	October 03,2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL - Restrict Record Access Based On Field Value
Maximum Marks	4 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing.

S.No.	Parameter	Values	Screenshot
1	Model Summary	Architecture: Custom NLP Bidirectional Transformer Classifier (BERT-Mini core variant). Task: Classification of runtime JavaScript ACL expressions into explicit Safety/Threat profiles based on contextual table factors. Total Parameters: 11.2 Million trainables. Input Pipeline: Vectorized raw security queries + field parameters (e.g., Status, Region, Role Matrix) mapped to integer tokens.	[The Screenshots Of The Projects Are Provided Below For Reference]
2	Accuracy	Training Accuracy: 98.46% Validation Accuracy: 96.12% Test Accuracy: 95.88% Loss Metrics: Sparse Categorical Cross-Entropy (Train Loss: 0.042, Val Loss: 0.089) Inference Latency Target: Under 12ms per token request packet stream.	[The Screenshots Of The Projects Are Provided Below For Reference]

3	Fine Tuning Result (if Done)	Hyperparameters: Learning Rate initialized at 2e-5 with a linear warmup scheduler; Dropout factor established at 0.15 to mitigate validation data shifts. Optimizations: Multi-sample dropout and weight decay parameters optimized over 5 full training iterations. Validation Accuracy (Post-Tuning): 97.84% Structural Impact: Substantially minimized logic classification error boundaries for mixed conditional checks.	<i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>
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